How to read a scientific study

Before you believe something, whether it is a new "miracle" supplement, a diet trick from a reel, or a headline that says "science has proved", it pays to tell a strong piece of evidence from a weak one. Here is a plain guide to finding your way in studies, spotting hype, and not being led by the nose. Practical, said straight.

Why you need a hierarchy of evidence

Not every "proof" weighs the same. Evidence-based medicine sits on a simple idea: judge a claim by the **quality and the type** of data behind it, not by how confidently it is said [5]. One tweet, one mouse study and one meta-analysis are not three equal opinions. They are three very different levels of certainty.

That is why there is a **pyramid (a hierarchy) of evidence**: the higher you go, the better the risk of being wrong is kept in check [1]. It is not dogma. A badly run RCT can be weaker than a well run observational study [2]. As a compass, though, it works very well. Hold this rule: the more serious the claim, the stronger the evidence you should want.

> **Main point:** Don't only ask "what did the study find". Ask **"what kind of study was it, and how well was it done"**. The strength of the conclusion follows the quality of the evidence, not the volume it is presented at [3,5].

The evidence pyramid: weakest to strongest

Walk it from the bottom up. Each step adds control over bias and over chance [1,2]:

1 Personal experience / anecdote

"It worked for me." One person, no control, plenty of bias. Useful as a prompt for a question. Never as proof.

2 Case report

A careful description of one case, or a few. It can flag a new phenomenon. It cannot prove a cause.

3 Observational study

It follows groups without an intervention (cohort, case-control). It finds **associations**, not necessarily a cause. See correlation vs. causation.

4 RCT (randomised controlled trial)

People are randomly assigned to groups (the intervention vs. a control or a placebo). Randomisation balances the confounders. That is why it is the gold standard for the question "does it work?" [1].

5 Systematic review

It gathers **all** the decent studies on a question, by a protocol set in advance. Not only the ones that fit the story [6].

6 Meta-analysis

It statistically combines the results of several studies into one estimate. The top floor, if the studies going in are good. Otherwise it is "garbage in, garbage out" [2,6].

Careful: A place on the standard pyramid always has to be corrected for **how well the study was done**. A modern approach (for example GRADE) rates certainty by risk of bias, consistency and the size of the effect, not only by the "label" of the study type [2,3].

Why "it worked for me" is not proof

Personal experience is worth a lot for motivation. As proof that something works, it is unreliable, however honestly you feel it. Three quiet traps:

- ✓ **Placebo effect.** Expecting to feel better can itself reduce perceived pain, fatigue and appetite. That is why a good RCT compares the intervention with a placebo. Without a control you don't know whether "the thing" worked, or the expectation [1].
- ✓ **Regression to the mean.** We reach for a supplement or a diet when we feel worst. The state then often improves on its own, simply by returning toward normal, and we wrongly credit the intervention.
- ✓ **Bias.** We remember the hits and forget the misses (confirmation bias). We look for what confirms what we already think. That is exactly what a study's method is built to hold down.

One person also has no control group. You never know what would have happened if you had **not** done it. A controlled study is what answers that, which is why one miracle story does not beat a well run experiment [5].

Correlation is not causation

This is the most common logical error in headlines. Two things moving together does not mean one causes the other. An observational study finds an **association**, and for good reason that is not enough:

- ✓ **A mixed-up cause (a confounder).** People who eat more fish tend to be healthier, but they also tend to exercise more, smoke less and have a higher income. Which of those is the cause?
- ✓ **The arrow can point the other way.** Does depression go with less physical activity because movement protects mood, or because depression takes away the energy to move? A correlation cannot tell them apart.

- ✓ **Chance.** If you compare enough things, some "association" shows up by chance alone [4].

A practical rule: when you read "X is linked with Y", add in your head **"linked, but it may not cause it"**. A conclusion about cause usually needs a randomised experiment [1].

In vitro, mice, and sample size

Before you read the conclusion, look at **who, and how many**. This is where most of the hype dies:

- ✓ **In vitro (in a test tube) and animal models.** What works on cells in a dish, or in mice, often **does not work in people**, or only at doses a person would never realistically take. It is a necessary first step in research. It is not proof for you.
- ✓ **Sample size.** A study on 12 people is a signal, not a conclusion. Small samples give unstable, easily exaggerated results, and they are harder to repeat [4].
- ✓ **Replication.** One "discovery" can be chance or a method error. Trust what has been confirmed repeatedly, across studies [4,6].

> **A fast filter:** If a headline shows a miraculous effect, find out whether it was people or mice/cells, and how many people. A lot of "breakthrough" news dies on that one question.

The p-value, effect size, and practical importance

"Statistically significant" sounds strong, but it means less than people think. A **p-value** says, roughly: how likely it would be to see a result this extreme (or more extreme) if the intervention actually **did nothing**. What a p-value is not:

- ✓ **It is not the probability that you are right.** $p < 0.05$ does not mean "95% certainty that it works". That is the most common myth about the p-value [7].
- ✓ **It is not the size or the importance of the effect.** With a big enough sample, a tiny, practically worthless difference still comes out "significant" [7].
- ✓ **The 0.05 threshold is a convention, not a law of nature.** $p = 0.049$ and $p = 0.051$ are the same thing in practice. Chasing the "star" produces distorted results [4,7].

So ask for the **effect size** (how much it actually changed) and for **practical importance**: "They lost 0.4 kg extra over a year compared with the control" can be statistically significant and still mean nothing for you. Always look for the **confidence interval** too. It shows the range the truth most likely sits in, and with it the uncertainty.

Conflicts of interest, predatory journals, and hype

A study does not appear in a vacuum. Before you trust a result, check the context:

- ✓ **Funding and conflicts of interest.** Is the study of a supplement paid for by the company that sells it? That does not have to be fraud, but it is a reason to be more careful. Look for the "funding" and "conflict of interest" section. A serious study states it.
- ✓ **Predatory journals.** There are "journals" that will print anything for a fee, with no real review. "It was published" is not enough on its own. What matters is **where**, and whether it went through peer review.
- ✓ **Media and social hype.** The headline exaggerates, drops the "in mice", hides the sample size and turns a correlation into a cause. Always try to find the **original study**, not only an article about an article about it.

A healthy reflex: the more sensational the claim, and the more someone earns from it, the harder the evidence you should demand before you pass it on [3,5].

Where to look, and a checklist before you believe it

Use decent sources instead of a random search: **PubMed** (a database of biomedical studies), **Cochrane** (systematic reviews of the highest quality) [6], and the classic handbook **"How to read a paper"** if you want to get better at the methods [8]. And before you believe something, or pass it on, run this checklist:

- ✓ **What type of study is it?** Anecdote, observational, RCT, meta-analysis. Where is it on the pyramid [1,2]?
- ✓ **People, or mice/cells?** And how many? A small sample = caution [4].
- ✓ **Correlation, or causation?** Was there a real intervention and a control group, or only observation?
- ✓ **How big is the effect in real life?** Not "it was significant", but "by how much, and for whom" [7].
- ✓ **Who paid, and where did it appear?** Conflict of interest, a peer-reviewed journal, or a predator?
- ✓ **Has it been confirmed more than once?** One result is a hypothesis. Agreement across studies is strength [6].
- ✓ **Does the headline match the study?** Find the original. The hype is usually in the translation into the media [3].

> **Action step:** Next time, before you share or act on a "science says", run just **three questions**: (1) what type of study, (2) people or mice, and how many, (3) how big is the effect in real life. Those three filters catch most of the nonsense before it reaches you. Be Effective!

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Sources: [1] OCEBM Levels of Evidence Working Group. The Oxford 2011 Levels of Evidence. Oxford Centre for Evidence-Based Medicine. [2] Murad MH, Asi N, Alsawas M, Alahdab F. New evidence pyramid. BMJ Evid Based Med. 2016;21(4):125-127. [3] Guyatt GH, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008;336(7650):924-926. [4] Ioannidis JPA. Why most published research findings are false. PLoS Med. 2005;2(8):e124. [5] Sackett DL, Rosenberg WM, Gray JA, Haynes RB, Richardson WS. Evidence based medicine: what it is and what it isn't. BMJ. 1996;312(7023):71-72. [6] Higgins JPT, Thomas J, et al. (eds). Cochrane Handbook for Systematic Reviews of Interventions. Cochrane, 2019. [7] Nuzzo R. Scientific method: statistical errors. Nature. 2014;506(7487):150-152. [8] Greenhalgh T. How to read a paper: the basics of evidence-based medicine. BMJ Books.